Aspects on Characteristic Classes

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Preface

This note is written for an informal summer 2026 seminar on characteristic classes. The goal is to explain several complementary viewpoints on the same invariant: a characteristic class as the pullback of a universal cohomology class on a classifying space; as an obstruction to the existence of global frames or non-vanishing sections; and as the Poincaré dual of a geometric zero or degeneracy locus [1, 2, 6].

Historically, characteristic classes grew out of the study of sphere bundles, tangent directions, and vector fields in the work of Stiefel and Whitney [4, 7, 8]. Chern introduced the characteristic classes of complex vector bundles, while Pontryagin developed characteristic cycles for differentiable manifolds [1, 3]. Thom later connected these ideas with transversality, cobordism, and the global topology of manifolds [5]. This note follows the modern language of vector bundles, principal bundles, universal bundles, obstruction theory, and intersection theory, with Milnor–Stasheff as the main reference [2].

Before we start the main text, we need to make the following convention on the base spaces:

1. Every space is assumed to be path-connected, locally path-connected (LPC), semi-locally simply connected (SLSC), paracompact, and Hausdorff, and maps are assumed to be continuous.
2. In all geometrical context, manifold M is assumed to be smooth.

1 Language of Vector Bundles and Sections

The goal of this section is to establish the fundamental theory that classifies (topological) vector bundles over paracompact Hausdorff spaces, more specifically, to prove the following classification theorem:

$$\mathrm{Vect}_n^{\mathbb{R}}(X) \cong [X, B\mathrm{O}(n)], \qquad \mathrm{Vect}_n^{\mathbb{C}}(X) \cong [X, B\mathrm{U}(n)],$$

and for oriented real vector bundles

$$\mathrm{Vect}_n^{\mathbb{R},+}(X) \cong [X, B\mathrm{SO}(n)],$$

where $[X, Y]$ is the homotopy classes of maps between spaces X and Y, and BG is the classifying space for principal G-bundles.

In other words, a principal G-bundle bundle $E \to X$ can be represented completely by a (homotopy class of) classifying map $f_E : X \to BG$, and in the following section, we will see that every characteristic class will be the pullback of the universal characteristic class via the classifying map:

$$\alpha_E := f_E^* \alpha_{\mathrm{univ}}.$$

We will not cover any characteristic class or classifying space in this section.

Vector Bundles and Basic Concepts. We may first recall the definition of a vector bundle, which is usually mentioned in the first course of differential geometry: Let $\mathbb{F}$ be a field ($\mathbb{R}$ or $\mathbb{C}$)

Definition 1.1 (Vector Bundles)**.** Let X be a space, a $\mathbb{F}$-vector bundle η on X with rank n is given by a sequence $F \hookrightarrow E \to X$ with the total space $E := E(\eta)$ and a surjection $\pi : E \to X$, known as the projection, and satisfies

1. For any $x \in X$, the preimage under projection $E_x := \pi^{-1}(x) \cong F$ is an n-dimensional $\mathbb{F}$-vector space called the fiber of η.
2. For each $x \in X$, there exists an open set $U \subseteq X$ contains x and a homeomorphism

$$\varphi_U : \pi^{-1}(U) \xrightarrow{\cong} U \times \mathbb{F}^n$$

such that $\mathrm{pr}_1 \circ \varphi|_U = \pi|_U$, and for each $y \in U$, the restriction $\varphi_U|_{E_y} : E_y \to \{y\} \times \mathbb{F}^n$ is an linear isomorphism.

The map φ_U in the second condition is called a "local trivialization", which can be summarized as the following diagram:

$$\begin{tikzcd} \pi^{-1}(U) \arrow[r, "\varphi_U", "\cong"'] \arrow[d, "\pi"'] & U \times \mathbb{F}^n \arrow[dl, "\mathrm{pr}_1"] \\ U & \end{tikzcd}$$

The $\mathbb{F}$-vector bundle $X \times \mathbb{F}^n$ over X is called the trivial bundle. In most cases in geometry, we consider the coefficient field to be $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, which

- If $\mathbb{F} = \mathbb{R}$, $E \to X$ is called a rank n real vector bundle.
- If $\mathbb{F} = \mathbb{C}$, $E \to X$ is called a rank n complex vector bundle.

Consider two $\mathbb{F}$-vector bundles η_0 and η_1 on the same base space X with both rank equals to n and projection π_i, $i = 0, 1$. One can defined the bundle isomorphism: Let $E_i := E(\eta_i)$ denotes the total space to two vector bundles, $i = 0, 1$

Definition 1.2 (Bundle Isomorphisms)**.** Vector bundles η_0 and η_1 are isomorphic if there exists an homeomorphism $f : E_0 \to E_1$ between total spaces such that the following diagram commutes:

$$\begin{tikzcd} E_0 \arrow[rr, "f"] \arrow[dr, "\pi_0"'] & & E_1 \arrow[dl, "\pi_1"] \\ & X & \end{tikzcd}$$

and for each $x \in X$, the restriction on fiber $f_x : (E_0)_x \to (E_1)_x$ is an $\mathbb{F}$-linear isomorphism.

We denote the isomorphism classes of $\mathbb{F}$-vector bundle over X with rank n as:

$$\mathrm{Vect}^{\mathbb{F}}_{n,\mathrm{iso}}(X) := \left\{\mathbb{F}\text{-vector bundles } E \to X \text{ with rank } n\right\}\Big/\text{Bundle isomorphisms.}$$

and the set of isomorphism classes of all $\mathbb{F}$-vector bundles in all ranks is

$$\mathrm{Vect}^{\mathbb{F}}_{\mathrm{iso}}(X) := \coprod_{n\in\mathbb{N}} \mathrm{Vect}^{\mathbb{F}}_{n}(X).$$

It is a central question to determine whether a vector bundle is trivial or not in the study of vector bundles, and to solve such a question, we need to study the sections of vector bundles, or vector fields:

Definition 1.3 (Section of Vector Bundles)**.** A section of a vector bundle $\pi : E \to B$ over X is a map $s : X \to E$ such that $\pi \circ \eta = \mathrm{id}_X$.

Proposition 1.4. *A rank n vector bundle is trivial if and only if it admits n nowhere dependent sections $s_1, \ldots, s_n$, i.e., n sections such that $s_1(x), \ldots, s_n(x)$ are linearly independent for all $x \in X$.*

One of the directions (trivial bundle admits n nowhere dependent sections) is trivial. For the converse direction, we need the following lemma:

Lemma 1.5. *Let η and ξ both be vector bundle over X, and $f : E(\eta) \to E(\xi)$ be a continuous function which maps each vector space that map each fiber $E(\eta)_x$ isomorphically to $E(\xi)_x$, then f is an isomorphism between vector bundles.*

Proof. For any point $x \in X$, take local trivialized open sets U and V of vector bundles η and ξ such that $x \in U \cap V$. Let $\varphi_\xi : \pi_\xi^{-1}(U) \to U \times \mathbb{R}^n$ and $\varphi_\eta : \pi_\eta^{-1}(V) \to V \times \mathbb{R}^n$ be local trivialization maps of η and ξ, then we need to show that the local coordinate form on $W := U \cap V$

$$\tilde{f} := \varphi_\xi^{-1} \circ f \circ \varphi_\eta : W \times \mathbb{R}^n \to W \times \mathbb{R}^n$$

is a homeomorphism. Since f maps the fiber $E(\eta)_x$ (linearly) isomorphically to $E(\xi)_x$, it has to satisfies $\pi_\xi \circ f(v) = \pi_\eta(v) = x$ for all $v \in E(\eta)_x$, i.e., $\forall x \in W$

$$\tilde{f}(x, v) = (x, A(x)v), \quad A(x) \in \mathrm{GL}_n(\mathbb{R}).$$

then the continuity of $\tilde{f}$ leads to the continuity of $A : W \to \mathrm{GL}_n(\mathbb{R})$, and we can state the inverse of $\tilde{f}$ as

$$\tilde{f}^{-1}(x, w) := (x, A(x)^{-1}w).$$

Since $A \mapsto A^{-1}$ is contentious map in $\mathrm{GL}_n(\mathbb{R})$, we know that $\tilde{f}^{-1}$ is also continuous. Finally, we may check that

$$\tilde{f}^{-1} \circ \tilde{f}(x, v) = \tilde{f}^{-1}(x, A(x)v) = (x, A(x)^{-1}A(x)v) = (x, v),$$

and similarly $\tilde{f} \circ \tilde{f}^{-1}(x, w) = (x, w)$. Thus, $\tilde{f}$ is a homeomorphism. Since the choice of points and locally trivializing sets is arbitrary, f is an isomorphism between vector bundles. $\square$

Now we can proof Proposition 1.4:

Proof of Proposition 1.4. ($\Rightarrow$) This direction is trivial.
($\Leftarrow$) Suppose the rank n vector bundle η with projection $\pi : E \to X$ admits n nowhere dependent sections

$$s_1, \ldots, s_n : X \to E$$

then, for all $x \in X$, $\{s_i(x) \mid i = 1, \ldots, n\}$ forms a basis for the fiber E_x. Then, consider the bundle map

$$F : X \times \mathbb{R}^n \to E, \quad (x, v) \mapsto \sum_{i=1}^{n} v_i s_i(x)$$

where $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$. Note that $F(x, v) = v_1 s_1(x) + \cdots + v_n s_n(x)$ is a vector in E_x, $\pi \circ F(x, v) = \mathrm{pr}_1(x, v) = x$ preserves fibers, and since

$$E_x := \mathrm{Span}\{s_i(x) \mid i = 1, \ldots, n\}$$

the restriction $F|_{\{x\}\times\mathbb{R}^n}$ is an linear isomorphism $\mathbb{R}^n \cong E_x$. By the continuity of sections s_i, $F : X \times \mathbb{R}^n \to E$ is also continuous. Since x can be chosen arbitrarily, apply the lemma, F is a bundle isomorphism. Thus, ξ is a trivial bundle. $\square$

Categorical Perspective of Vector Bundles. More generally, if there are two $\mathbb{F}$-vector bundles η_0 and η_1 on two different base spaces X_0 and X_1, with total spaces denotes by $E_i := E(\eta_i)$ and projections

$$\pi_0 : E_0 \to X_0, \qquad \pi_1 : E_1 \to X_1.$$

One can define the bundle map between them:

Definition 1.6 (Bundle Maps)**.** A map $\hat{f} : E_0 \to E_1$ is a bundle map if there exists a map $f : X_0 \to X_1$ such that

1. $f \circ \pi_0 = \pi_1 \circ \hat{f}$, i.e., the following diagram commutes:

$$\begin{tikzcd} E_0 \arrow[r, "\hat{f}"] \arrow[d, "\pi_0"'] & E_1 \arrow[d, "\pi_1"] \\ X_0 \arrow[r, "f"] & X_1 \end{tikzcd}$$

2. and, for any $x \in X_0$, the fiberwise restriction $\hat{f}_x : (E_0)_x \to (E_1)_{f(x)}$ is a linear map.

Here, the map $\hat{f} : E_0 \to E_1$ is called a bundle map covering $f : X_0 \to X_1$, and we denote the collection of bundle maps covering f as

$$\mathrm{Bun}_f(\eta_0, \eta_1) := \Big\{\text{Bundle maps } \hat{f} : E_0 \to E_1 \text{ that covering } f : X_0 \to X_1\Big\}.$$

Equivalently, given a map $f : X \to Y$ and a vector bundle η with projection $\pi_Y : E \to Y$, one can defined the pullback bundle on X:

Definition 1.7 (Pullback Bundle)**.** Let $f : X \to Y$ be a map, and $\pi_Y : E \to Y$ gives a vector bundle ξ on X. Then the total space of the pullback bundle $f^*\xi$ on X is defined by

$$f^*E := \{(x, e) \in X \times E \mid f(x) = \pi(e)\}$$

and the bundle projection $\pi_X : f^*E \to Y$ is just the natural projection on product, given by $\pi(x, e) = x$. The fiber is given by $(f^*E)_x \cong E_{f(x)}$ for any $x \in X$.

A direct fact is that for any vector bundle η on X given by $\pi : E \to B$, the pullback $\mathrm{id}_X^* \eta = \eta$ since $\mathrm{id}(x) = x = \pi(e)$ is just a tautology.
Note that one can define the bundle map $\hat{f} : f^*E \to E$ covering $f : X \to Y$ given by $\hat{f}(b, e) = e$, and this construction gives an isomorphism of the set of bundle maps for any vector bundle η on X:

Proposition 1.8. *Let X and Y be spaces, and $f : X \to Y$ is a continuous map. For $\mathbb{F}$-vector bundle η on X and ξ on Y, there is isomorphism $\mathrm{Bun}_f(\eta, \xi) \cong \mathrm{Bun}_{\mathrm{id}_X}(\eta, f^*\xi)$.*

Proof. Let η and ξ be vector bundles on X and Y with the corresponding projection be $\pi_X : E(\eta) \to X$ and $\pi_Y : E(\xi) \to Y$. Consider the bundle map $\hat{f} = \mathrm{pr}_2 : f^*E \to E$ covering f defined by $\hat{f}(x, e) = e$. The idea is simply construct the bundle map $\tilde{\varphi} \in \mathrm{Bun}_{\mathrm{id}_X}(\eta, f^*\xi)$ corresponding to $\varphi \in \mathrm{Bun}_f(\eta, \xi)$ such that the following diagram commutes:

$$(*) \qquad \begin{tikzcd} E(\eta) \arrow[rr, "\tilde{\varphi}"] \arrow[dr, "\pi_X"'] & & f^*E(\xi) \arrow[r, "\hat{f}"] \arrow[dl, "\mathrm{pr}_1"] & E(\xi) \arrow[d, "\pi_Y"] \\ & X \arrow[rr, "f"] & & Y \end{tikzcd} \iff \begin{tikzcd} E(\eta) \arrow[r, "\varphi"] \arrow[d, "\pi_X"'] & E(\xi) \arrow[d, "\pi_Y"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

and show that this is a bijective correspondence.
For any bundle map $\varphi \in \mathrm{Bun}_f(\eta, \xi)$ covering f, since the definition of bundle map requires $\pi_Y \circ \varphi = f \circ \pi_X$, one can defined another bundle map $\tilde{\varphi} \in \mathrm{Bun}_{\mathrm{id}_X}(\eta, f^*\xi)$ given by

$$\tilde{\varphi}(e) = (\pi_X(e), \varphi(e)) \qquad \forall e \in E(\eta).$$

It is not hard to check that this map ensures the diagram $(*)$ commutes.
Conversely, for a bundle map $\psi \in \mathrm{Bun}_{\mathrm{id}_X}(\eta, f^*\xi)$, consider the bundle map $\hat{f} \circ \psi \in \mathrm{Bun}_f(\eta, \xi)$. By the definition of $\hat{f}$, we have

$$\pi_Y \circ (\hat{f} \circ \psi) = f \circ \mathrm{pr}_1 \circ \psi,$$

and since ψ is a bundle map, $\mathrm{pr}_1 \circ \psi = \pi_X$. Thus, $\pi_Y \circ (\hat{f} \circ \psi) = f \circ \pi_X$, i.e., $\hat{f} \circ \psi$ is a bundle map.
Finally, we shall check that these two constructs are the inverse of each other: For any $\varphi \in \mathrm{Bun}_f(\eta, \xi)$,

$$\hat{f} \circ \tilde{\varphi}(e) = \hat{f}(\pi_X(e), \varphi(e)) = \varphi(e) \quad \forall e \in E(\eta),$$

and for any $\psi \in \mathrm{Bun}_{\mathrm{id}_X}(\eta, f^*\xi)$, since ψ covering id_X, $\pi_X(e) = \mathrm{pr}_1 \circ \psi(e)$

$$(\pi_X(e), \hat{f} \circ \psi(e)) = (\mathrm{pr}_1 \circ \psi(e), \mathrm{pr}_2 \circ \psi(e)) = \psi(e).$$

Thus, these two constructions are the inverse of each other, which shows $\mathrm{Bun}_f(\eta, \xi) \cong \mathrm{Bun}_{\mathrm{id}_X}(\eta, f^*\xi)$. $\square$

The categorical understanding of the isomorphism in Proposition 1.8 is given in the following remark:
Remark. It is not really rigorous that the pullback bundle admits "exactly the same information" as the bundle map. Instead, let $\mathsf{Vbl}_{\mathbb{F}}(X)$ denote the category of $\mathbb{F}$-vector bundles over X and $\mathsf{Vbl}_{\mathbb{F}}$ be the category of $\mathbb{F}$-vector bundles (without specifying the base space), both take bundle maps as morphisms. Then the isomorphism gives a natural isomorphism of Hom-functors in Vbl:

$$\mathrm{Bun}_f(-, \xi) \xrightarrow{\cong} \mathrm{Bun}_{\mathrm{id}_X}(-, f^*\xi),$$

and furthermore, let η and ξ be vector bundle over X and Y, the $\mathrm{Hom}_{\mathsf{Vbl}}$ set is given by

$$\begin{aligned} \mathrm{Hom}_{\mathsf{Vbl}}(\eta, \xi) &= \coprod_{f : X \to Y} \mathrm{Bun}_f(\eta, \xi) \cong \coprod_{f : X \to Y} \mathrm{Bun}_{\mathrm{id}_X}(\eta, f^*\xi) \\ &= \coprod_{f : X \to Y} \mathrm{Hom}_{\mathsf{Vbl}_{\mathbb{F}}(X)}(\eta, f^*\xi). \end{aligned}$$

This shows that the morphism $\varphi \in \mathrm{Hom}_{\mathsf{Vbl}_{\mathbb{F}}}(\eta, \xi)$ can be written as a pair $(f, \tilde{\varphi})$, where $f \in \mathrm{Hom}_{\mathsf{Top}}(X, Y)$ and $\tilde{\varphi} \in \mathrm{Hom}_{\mathsf{Vbl}_{\mathbb{F}}(X)}(\eta, f^*\xi)$. Thus, the projection functor

$$\mathrm{Pr} : \mathsf{Vbl}_{\mathbb{F}} \to \mathsf{Top}$$

defined by sending each vector bundle η to its base space gives an Grothendieck fibration with fibers $\mathrm{Pr}^{-1}(X) = \mathsf{Vbl}_{\mathbb{F}}(X)$.

The most remarkable property of the pullback bundle is that it is functorial; more specifically, it satisfies the following proposition:

Proposition 1.9 (Functoriality of Pullback Vector Bundles)**.** *Let $\pi : E \to Z$ gives a vector bundle η over Z, and $X \xrightarrow{f} Y \xrightarrow{g} Z$ be continuous maps. Then there is a canonical isomorphism between pullback vector bundles $f^*g^*\eta \cong (g \circ f)^*\eta$.*

Proof. By the definition of pullback bundles, the total space of $g^*\eta$ is given by

$$g^*E = \{(y, e) \in Y \times E \mid g(y) = \pi(e)\}.$$

Then, pullback this total space again using $f : X \to Y$, we can obtain

$$f^*(g^*E) := \big\{(x, (y, e)) \in X \times g^*E \mid f(x) = y\big\}.$$

Combine these two definitions, since g^*E require $g(y) = \pi(e)$,

$$f^*(g^*E) = \{(x, e) \in X \times E \mid (g \circ f)(x) = \pi(e)\} = (g \circ f)^*E.$$

The explicit isomorphism is canonically given by $\big(x, (f(x), e)\big) \mapsto (x, e)$ and the bijectivity is simply given by the fact that f is a map, and there exists a unique image $f(x) \in Y$ for any $x \in X$ under f. $\square$

The functoriality we proved here in Proposition 1.9 leads to the following categorical definition of the category of vector bundles over space X: Let Top be the category of topological spaces, and Grpd be the category of groupoids,

Definition 1.10 (Moduli Functor of Vector Bundles)**.** The collection of all $\mathbb{F}$-vector bundles with rank n defines a cotravariant pseudofunctor

$$\mathrm{Vect}_n^{\mathbb{F}}(-) : \mathsf{Top}^{\mathrm{op}} \to \mathsf{Grpd},$$

defined by sending each space $X \mapsto \mathrm{Vect}_n^{\mathbb{F}}(X)$ to the groupoid given as follows:

- $\mathrm{ob}\,\mathrm{Vect}_n^{\mathbb{F}}(X) :=$ All $\mathbb{F}$-vector bundles η with rank n over X.
- $\mathrm{Hom}_{\mathrm{Vect}_n^{\mathbb{F}}(X)}(\eta, \xi) :=$ Vector bundle isomorphisms between η and ξ.

and the functor sends continuous map $f : X \to Y$ to the pullback functor $f^* : \mathrm{Vect}_n^{\mathbb{F}}(Y) \to \mathrm{Vect}_n^{\mathbb{F}}(X)$ between groupoids.

This is a pseudofunctor since for $X \xrightarrow{f} Y \xrightarrow{g} Z$ the functor does not really satisfies $f^*g^* = (g \circ f)^*$. Instead, Proposition 1.9 gives a natural isomorphism $\alpha_{f,g,\eta} : f^*g^*\eta \Rightarrow (g \circ f)^*\eta$ such that the following two diagrams (naturality and associative coherence) commutes: For bundle isomorphism $\varphi : \eta \to \xi$ in $\mathrm{Vect}^{\mathbb{F}}(X)$ and continuous maps $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} K$

$$\begin{tikzcd} f^*g^*\eta \arrow[r, "\alpha_{f,g,\eta}"] \arrow[d, "f^*g^*\varphi"'] & (g \circ f)^*\eta \arrow[d, "(g\circ f)^*\varphi"] \\ f^*g^*\xi \arrow[r, "\alpha_{f,g,\xi}"] & (g \circ f)^*\xi \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} f^*g^*h^* \arrow[r, "f^*\alpha_{g,h}"] \arrow[d, "\alpha_{f,g}h^*"'] & f^*(h \circ g)^* \arrow[d, "\alpha_{f,h\circ g}"] \\ (g \circ f)^*h^* \arrow[r, "\alpha_{g\circ f,h}"] & (h \circ g \circ f)^* \end{tikzcd}.$$

Note that the set of isomorphism classes we mentioned before is actually given by a Set-valued functor $\mathrm{Vect}^{\mathbb{F}}_{n,\mathrm{iso}}(-) : \mathsf{Top}^{\mathrm{op}} \to \mathsf{Set}$, where $\mathrm{Vect}^{\mathbb{F}}_{n,\mathrm{iso}}(-) := \pi_0 \mathrm{Vect}^{\mathbb{F}}_n(-)$ is the set of connected components in the groupoid. Similarly, we can define the functor

$$\mathrm{Vect}^{\mathbb{F}}_{\mathrm{iso}}(-) := \pi_0 \mathrm{Vect}^{\mathbb{F}}(-) = \coprod_{n\in\mathbb{N}} \mathrm{Vect}^{\mathbb{F}}_{n,\mathrm{iso}}(-) : \mathsf{Top}^{\mathrm{op}} \to \mathsf{Set}$$

which provides all isomorphism classes of $\mathbb{F}$-vector bundles.

Remark (Sheaf-Theoretic Viewpoint). We can also use sheaf theoretic language to study the moduli problem of vector bundles: Let X be a fixed space. Restricting he functor $\mathrm{Vect}^{\mathbb{F}}_n(-)$ to the category of open subsets of X gives a presheaf of groupoids

$$\mathcal{V}^{\mathbb{F}}_{n,X} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathsf{Grpd}$$

defined by $U \mapsto \mathrm{Vect}^{\mathbb{F}}_n(U)$. For an inclusion $i : V \hookrightarrow U$, the associated functor

$$\mathrm{res} : \mathrm{Vect}^{\mathbb{F}}_n(U) \to \mathrm{Vect}^{\mathbb{F}}_n(V)$$

is given by restriction of vector bundles $\eta \mapsto \eta|_V := i^*\eta$. In fact, this presheaf of groupoids is a stack with respect to the open-cover topology, since vector bundles and their isomorphisms satisfy descent. In these notes, however, we will not introduce the language of moduli stacks.

Application of Partition of Unity on Base Space. In the final part of this section, we will explain why we need "sufficiently good" spaces as a base space. More specifically, we need the paracompactness and Hausdorffness to ensure there exists a simple way to generalize local property to global: Partition of Unity.

Definition 1.11 (Partition of unity)**.** Let X be a topological space and $\mathcal{U} := \{U_\alpha\}_{\alpha\in A}$ an open cover of X. A partition of unity corresponding to $\mathcal{U}$ is a family of continuous functions $\{\rho_\alpha\}_{\alpha\in A} \subseteq C(X,[0,1])$ such that:

1. the family $\{\rho_\alpha\}_{\alpha\in A}$ is locally finite, i.e., for any $\alpha \in A$ and $x \in X$, there exists an neighborhood $N_x \ni x$ that only intersects with finite may supports $\mathrm{supp}(\rho_\alpha)$;
2. $\mathrm{supp}(\rho_\alpha) \subseteq U_\alpha$ for all $\alpha \in A$;
3. for every $x \in X$ one has
$$\sum_{\alpha\in A} \rho_\alpha(x) = 1,$$
where the sum is well-defined by local finiteness.

Theorem 1.12 (Existence of partitions of unity)**.** *Let X be a paracompact Hausdorff space and $\mathcal{U} := \{U_\alpha\}_{\alpha\in A}$ an open cover of X. Then there exists a partition of unity $\{\rho_\alpha\}_{\alpha\in A}$ corresponding to $\mathcal{U}$.*

This is a classical result of point set topology, and we may use the theorem without proof.

The most important application of the partition of unity is to prove the existence of a bundle metric (or Riemannian metric) on any vector bundle. We would expect such a structure to exist partially due to Urysohns metrization theorem, since paracompact Hausdorff spaces are normal, and thus, metrizable.

Theorem 1.13. *Every finite-rank topological real (resp. complex) vector bundle $\pi : E \to X$ admits a bundle metric, i.e., a fiberwise defined inner (resp. Hermite) product*

$$\langle -,- \rangle_x : E_x \times E_x \to \mathbb{R}$$

that is continuously defined for all $x \in X$.

If the base space is a smooth or complex manifold, we can safely consider the same structure that is smoothly or holomorphically defined on any point $x \in X$.

Proof of Theorem 1.13. Consider the local trivialization cover $\mathcal{U} := \{U_i\}_{i\in I}$ of $E \to X$ with trivialization map denote as $\varphi_i : E|_{U_i} \to U_i \times \mathbb{F}^r$, and let $\{\rho_i\}_{i\in I}$ be the partition of unity associate to the cover $\mathcal{U}$. Let h_i be standard Euclidean (resp. Hermite) metric on $\varphi_i(E|_{U_i})$, and we may consider the globally defined metric

$$h(-,-)_x := \sum_i \rho_i(x)h_i(-,-),$$

which is indeed a bundle metric. □

A special case of this formalism is the bundle metric on the tangent bundle of a smooth manifold, which is called the Riemannian metric.

An application of this theorem is to study the short exact sequence of vector bundles and splitting.

Definition 1.14 (Short Exact Sequence)**.** Let X be a sufficiently good space, a short exact sequence of vector bundles over X

$$0 \longrightarrow E_0 \xhookrightarrow{i} E \xrightarrow{p} E_1 \longrightarrow 0$$

is defined by a fiberwise short exact sequence of vector spaces:

$$0 \longrightarrow (E_0)_x \xhookrightarrow{i_x} E_x \xrightarrow{p_x} (E_1)_x \longrightarrow 0$$

fo any $x \in X$, where i and p are bundle maps. Equivalently, we can define the short exact sequence of a vector bundle as the stalkwise short exact sequence of sheaves.

Here, we use the total space of the vector bundle to represent the bundle since the choice of base space is self-evident.

Since finite-dimensional vector spaces are freely generated by some finite set, due to the universal property of free objects, the short exact sequence of vector spaces always splits. This fact leads to the very first result of the splitting of the vector bundle short exact sequence:

Proposition 1.15. *Any short exact sequence of vector bundles*

$$0 \longrightarrow E_0 \xhookrightarrow{i} E \xrightarrow{p} E_1 \longrightarrow 0$$

locally splits. That is, for any point $x \in X$, one can find sufficiently small open neighborhood $U \subseteq X$ such that

$$E|_U \cong E_0|_U \oplus E_1|_U.$$

Proof. By local trivialization, in locally trivializing an open cover, the short exact sequence can be transformed into the following sequence:

$$0 \longrightarrow U \times \mathbb{F}^{r_1} \xhookrightarrow{(\mathrm{id},i)} U \times \mathbb{F}^r \xrightarrow{(\mathrm{id},p)} U \times \mathbb{F}^{r_2} \longrightarrow 0.$$

And this is simply a short exact sequence of vector spaces, which always splits. □

However, the global result needs more technique, since in general, local splitting may not glue to global splitting.

Theorem 1.16. *Let X be a sufficiently good space, the short exact sequence of a vector bundle*

$$0 \longrightarrow E_0 \xhookrightarrow{i} E \xrightarrow{p} E_1 \longrightarrow 0$$

always splits, i.e., E can be written in Whitney sum

$$E \cong E_0 \oplus E_1.$$

Proof. By Theorem 1.13, we may choose a bundle metric h on E. Then, by the injective map $i : E_0 \hookrightarrow E$, one can fiberwisely define the orthogonal complement $(E_0)^\perp$ of E_0 as a subbundle of E. By the exactness of the sequence, $\ker(p) = \operatorname{im}(i)$, i.e., $p|_{(E_0)^\perp} : (E_0)^\perp \to E_1$ is an linear isomorphism. Thus, the natural decomposition

$$E \cong E_0 \oplus (E_0)^\perp \cong W_0 \oplus E_1.$$

This is exactly what we want to show. $\square$

As one of the most important application of partition of unity, an essential fact is that the functor $\mathrm{Vect}^{\mathbb{F}}(-)$ is homotopically invariant.

Theorem 1.17 (Homotopy Invariance of Vector Bundles)**.** *Let $f_0 \simeq f_1 : X \to Y$ be homotopic continuous maps, and $E \to Y$ be some finite-rank vectoe bundle, then $f_0^* E = f_1^* E$.*

To prove the theorem, we need a lemma:

Lemma 1.18 (Cylinder Lemma)**.** *Let $\pi : E \to X \times I$ be a vector bundle on $I \times X$, and $\mathrm{pr}_1 : X \times I \twoheadrightarrow X$ be the projection. Then, we have the isomorphism $E \cong \mathrm{pr}_1^*(i_0^* E)$.*

Proof. We may begin with the following intuitive two claims:
Claim 1: For any $x \in X$, there exists an open neighborhood $U \subseteq X$ of x that $E|_{U\times I} \cong U \times \mathbb{R}^r$ is trivial.
Claim 2: A local trivialization $\varphi : E|_U \to U \times \mathbb{F}^r$ leads to fiber isomorphism $E_x \cong E_y$ for any points $x, y \in U$.
To prove the proposition, note that for any $t \in I$, the local trivialization leads to product neighborhood $U_t \times J_t \subseteq X \times I$ containing (x, t), such that $E|_{U_t\times J_t}$ is trivial. By the compactness of the unit closed interval I, one can find finite may $t \in I$ such that $\{J_{t_i}\}_{i=1,\dots,N}$ covers I, and we shall denote $J_i := J_{t_i}$ and $U_i := U_{t_i}$ for the following parts of the proof. Let

$$U := U_1 \cap U_2 \cap \cdots \cap U_N.$$

Then, E trivializes on each $V \times J_i$. One can take partition

$$0 = t_0 < t_1 < \cdots < t_m = 1,$$

such that for each index j, the interval $[t_j, t_{j+1}]$ contains in some $J_{\nu(i)}$. This construction ensures that E trivializes on each $U \times [t_j, t_{j+1}]$. To prove the bundle trivializes on $U \times I$, it remains to show that if E trivialize on $U \times [a, b]$ and $U \times [b, c]$ by φ_- and φ_+, then the local trivialization glues into a trivialization φ on $U \times [a, c]$. More precisely, consider

$$\varphi_\pm(e) := (\pi(e), \widehat{\varphi}(e)), \qquad \widehat{\varphi}(e) \in \mathbb{F}^r$$

One can consider the transition function given by

$$A : U \to \mathrm{GL}_r(\mathbb{F}), \qquad y \mapsto A(y) := (\widehat{\varphi}_+ \circ \varphi_-^{-1})|_{\pi^{-1}(y)}$$

Then we may redefined the trivialization as

$$\tilde{\varphi}_+(e) := ((y, t), A(y)^{-1}\widehat{\varphi}_+(e)), \qquad e \in \pi^{-1}(y \times [b, c]).$$

This is still a local trivialization, and since $\varphi_-(e) = \big((y, c), A(y)^{-1}\widehat{\varphi}_+(e)\big) = \tilde{\varphi}_+(e)$ for any $e \in \pi^{-1}(y, c)$, we obtain an local trivialization on $V \times [a, c]$. By finite steps of induction, it is simple to prove the first claim.
To see the second claim, consider the local trivialization that $E|_U \cong U \times F^r$. Restriction of the trivialization to points x and y gives

$$E_x \cong \{x\} \times F^r \cong \{y\} \times \mathbb{F}^r \cong E_y.$$

This proves the second claim.

With the first claim, one may choose open cover $\mathcal{U} := \{U_\alpha\}_{\alpha\in A}$ of X such that $E|_{U_\alpha\times I}$ is trivial with local trivialization

$$\Phi_\alpha : E|_{U_\alpha\times I} \xrightarrow{\cong} U_\alpha \times I \times \mathbb{F}^r.$$

By Theorem 1.12, we can take a partition of unity $\{\lambda_\alpha : X \to [0,1]\}_{\alpha\in a}$ subordinate to $\mathcal{U}$, and consider a total order $\prec$ on A. We may use the second claim that a local trivialization

$$\Phi_\alpha : E|_{U_\alpha\times I} \to U_\alpha \times I \times \mathbb{F}^r, \qquad e \mapsto (\pi(e), \phi_{\alpha,x,t}(e)) \text{ for } (x,t) := \pi(e)$$

induces continuously the fiber isomorphism $P_\alpha(x;t,s) : E_{(x,t)} \xrightarrow{\cong} E_{(x,s)}$ for any $(x,t),(x,s) \in U \times I$. Fix $x_0 \in X$, by locally finiteness, one can find an open neighborhood W of x_0 and finite index set

$$F_W := \{\alpha_1 \prec \alpha_2 \prec \cdots \prec \alpha_N\}$$

such that $W \cap \operatorname{supp}\lambda_{\alpha_i} \neq \varnothing$. Defined $s_0(x,t) = 0$ and

$$s_j(x,t) := t\sum_{i=1}^{\alpha_j} \lambda_{\alpha_i}(x), \qquad 1 \le j \le N.$$

Note that by the definition of partition of unity, $s_N(x,t) = 1$. Define the base maps

$$g_j : W \times I \to X \times I, \qquad (x,t) \mapsto (x, s_j(x,t)),$$

in particular, $g_0(x,t) = (x,0) = (i_0 \circ \mathrm{pr}_1|_{W\times I})(x,t)$ and $g_N(x,t) = (x,t)$. Thus, let $\mathrm{pr}_1^W := \mathrm{pr}_1|_{W\times I}$

$$g_0^*E \cong (\mathrm{pr}_1^W)^* i_0^* E, \qquad g_N^*E \cong E|_{W\times I}.$$

Note that on point (x,t),

$$(g_{j-1}^*E)_{x,t} = E_{(x,s_{j-1}(x,t))}, \qquad (g_j^*E)_{x,t} = E_{(x,s_j(x,t))}.$$

For $j = 1,\dots,N$, apply the local trivialization Φ_{α_i} gives the transportation, we may defined the step map $R_j^W : g_{j-1}^*E \to g_j^*E$ as

$$R_j^W((x,t),e) := \begin{cases} \Big((x,t), P_{\alpha_j}(x; s_{j-1}(x,t), s_j(x,t))e\Big), & \text{if } (x,t) \in (W \cap U_{\alpha_j}) \times I \\ \Big((x,t), e\Big), & \text{if } (x,t) \in (W \setminus \operatorname{supp}\lambda_{\alpha_j}) \times I. \end{cases}$$

By the fact that $\operatorname{supp}\lambda_{\alpha_j} \subseteq U_{\alpha_J}$, the step map R_j^W is globally and continuously defined on M, and well-defined on the overlap. We may write the map

$$\Theta_W := R_N^W \circ \cdots \circ R_1^W : (\mathrm{pr}_1^W)^* i_0^* E \to E|_{W\times I}$$

and since R_j^W is fiberwisely given by the fiber isomorphism induced by local trivialization, Θ_W and R_j^W are bundle isomorphisms.

It remains to show that the map glues to a global isomorphism. Consider open sets W and W' with finite index sets F_W and $F_{W'}$. Let $Z := W \cap W'$, and $F := F_W \cup F_{W'}$, since F is still a finite set, we can write it in the total order

$$F := \{\gamma_1 \prec \cdots \prec \gamma_M\}.$$

One can repeat the previous construction on Z using F as the index set and obtain

$$\Theta_Z^F : (\mathrm{pr}_1^Z)^* i_0^* E \to E|_{Z\times I}.$$

By restricting the induces, if $\gamma \in F \setminus F_W$, then $W \cap \operatorname{supp} \lambda_\gamma = \varnothing$, and thus, every elements indexed in $F \setminus F_W$ induces identity maps, similar statement hold for $F \setminus F_{W'}$. Simply omit these identity, one can obtain that

$$\Theta_W|_{Z\times I} = \Theta_Z^F = \Theta_{W'}|_{Z\times I}.$$

This shows that the fact that Θ_W and $\Theta_{W'}$ coincides on overleaps, ans thus, can glue together. In this way, our local construction uniquely glues to an bundle isomorphism

$$\Theta : \mathrm{pr}_1^* i_0^* E \xrightarrow{\cong} E.$$

This is exactly what we have stated in the proposition. □

With the cylinder lemma, we may now prove the homotopy invariant theorem:

Proof of Theorem 1.17. Let $H : I \times X \to Y$ give homotopy $f_0 \simeq f_1$, $f_t := H(-,t)$. Then pullback the vector bundle η on Y given by $E \to Y$ using H^*, one can obtain $H^*\eta$, with projection $H^*E \to I \times X$. Let the inclusion map be

$$i_t : X \hookrightarrow I \times X, \quad i_t(x) = (x,t) \ \forall t \in [0,1].$$

Then $f_t^* = i_t^* H^*$. Thus, it is sufficient to show that

$$i_1^*(H^*E) \cong i_0^*(H^*E).$$

Consider the projection $\mathrm{pr}_1 : I \times X \twoheadrightarrow X$. Note that the cylinder lemma tells us for any vector bundle $\mathcal{E} = H^*E \to I \times X$,

$$\mathcal{E} \cong \mathrm{pr}_1^*(i_0^* E)$$

By pullback the isomorphism above again by i_1, with the fact the $\mathrm{pr}_1 \circ i_t = \mathrm{id}_X$ for all $t \in I$

$$i_1^*\mathcal{E} \cong i_1^*(\mathrm{pr}_1^* i_0^*\mathcal{E}) \cong (\mathrm{pr}_1 \circ i_1)^* i_0^*\mathcal{E} \cong i_0^*\mathcal{E},$$

i.e., $f_0^*E \cong f_1^*E$, which completes the proof of the theorem. □

In particular, let X and Y be spaces, $f : X \to Y$ be a homotopy equivalence. Then the pullback gives an isomorphism

$$f^* : \mathrm{Vect}_r^{\mathbb{F}}(Y) \xrightarrow{\cong} \mathrm{Vect}_r^{\mathbb{F}}(X).$$

An simple corollary is given below:

Corollary 1.19. *Any vector bundle on contractable spaces is trivial.*

2 Vector Bundles as Sheaves

Instead of discussing the moduli problem of vector bundles, we can use the language of sheaves to study specific vector bundles, and this perspective will lead to the central philosophy of the study of bundles on manifolds: The study of transition maps. Before we dive into vector bundles as sheaves, let us review the basic notion of a sheaf.

The philosophy of sheaf is to describe a certain structure that is locally defined on a space, and behaves well under gluing operations. Let X be some space, the category of open sets $\mathrm{Open}(X)$ is given by

- The object is $\mathrm{ob}\,\mathrm{Open}(X) = \mathcal{T}_X$ the topology of X, i.e., the collection of all open sets $\mathrm{U} \hookrightarrow X$.
- The morphism is given by inclusions of open sets $V \hookrightarrow U$ in X.

Definition 2.1 (Presheaf)**.** Let X be a space and C be some subcategory of Set, a C-valued presheaf over X is a contravariant functor

$$\mathcal{F} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathsf{C}$$

Explicitly, a presheaf consists of the following data:

- For each open set $U \in \mathrm{Open}(X)$ an object $\mathcal{F}(U) \in \mathsf{C}$, also denotes as $\Gamma(U, \mathcal{F}) := \mathcal{F}(U)$. Since $\mathcal{F}(U)$ is often a set, the elements $s \in \mathcal{F}(U)$ are called a section of $\mathcal{F}$.
- For any inclusion $V \hookrightarrow U$, a restriction map $\mathrm{res}_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$, which often written as $s \mapsto s|_V$ on sections.

and it needs to satisfy the following axioms:

1. (Identity condition) $\mathrm{res}_{UU} = \mathrm{id}_{\mathcal{F}(U)}$.
2. (Composition law / Functoriality) For any $V \hookrightarrow U \hookrightarrow W$, $\mathrm{res}_{WV} = \mathrm{res}_{UV} \circ \mathrm{res}_{WU}$.

We often take C to be some Abelian category, for example, Ab, cRing, or R-Mod. The morphism $\varphi : \mathcal{F}_1 \to \mathcal{F}_2$ between presheaves, of course, should be a natural transformation:

Definition 2.2 (Morphism of Presheaves)**.** A morphism $\varphi : \mathcal{F}_1 \to \mathcal{F}_2$ between C-valued presheaves is a collection of morphisms φ_U (for all $U \subseteq X$ open) in C such that the following natural square commutes: For all $U, V \subseteq X$ open, $V \hookrightarrow U$

$$\begin{tikzcd}
\mathcal{F}_1(U) \arrow[r, "\varphi_U"] \arrow[d, "\mathrm{res}_{UV}"'] & \mathcal{F}_2(U) \arrow[d, "\mathrm{res}_{UV}"] \\
\mathcal{F}_1(V) \arrow[r, "\varphi_V"] & \mathcal{F}_2(V)
\end{tikzcd}$$

and the composition of morphisms is defined in a more obvious way that $(\varphi \circ \psi)_U := \varphi_U \circ \psi_U$.

In this way, we obtain the category of C-valued presheaves $\mathsf{PSh}(X, \mathsf{C})$ over X. Here are some examples of presheaves:

Example (Constant Presheaf)**.** Let A be an abelian group, let $\mathcal{A} : \mathrm{Open}(X) \to \mathsf{Ab}$ be a presheaf on X such that for any open set $U \subseteq X$, $\mathcal{A}(U) = A$, and all restrictions are the identity map id_A. It is not hard to check that this is indeed a presheaf.

and a more nontrivial one

Example (Presheaf of Continuous Functions)**.** Let X be some space, $\mathcal{C}_X^0 : \mathrm{Open}(X)^{\mathrm{op}} \to \mathsf{cRing}$ be a presheaf defined by for any open set $U \subseteq X$

$$\mathcal{C}_X^0(U) := \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$$

and for $V \hookrightarrow U$ the restriction $\mathrm{res}_{UV}(f) := f|_V$ is just the ordinary restriction. Then $\mathcal{C}_X^0$ is a presheaf of commutative rings. Similarly, one can define the presheaf of C^k-functions $\mathcal{C}_X^k$ and the presheaf of C^∞-functions $\mathcal{C}_X^\infty$.

Presheaves only require the restriction in open sets. To consider a further construction of gluing open sets, we can upgrade presheaves to sheaves:

Definition 2.3 (Sheaf)**.** A presheaf $\mathcal{F} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathsf{C}$ is a sheaf if for any open set $U \subseteq X$ and open cover $\mathcal{U} := \{U_i\}_{\in I}$ of X, the following condition holds: Given compatible local sections $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$. Then, there exists unique $s \in \mathcal{F}(U)$ such that for any $i \in I$, $s|_{U_i} = s_i \in \mathcal{F}(U_i)$.

The category of C-valued sheaves over X with morphism just presheaf morphisms is denoted by $\mathsf{Sh}(X, \mathsf{C})$, and we may make the convention that for a sheaf $\mathcal{F}$ on X and an open set $V \subseteq X$, $\mathcal{F}|_V$ is the restriction of the sheaf $\mathcal{F}$ to V. Note that a more general notion of sites and sheaves on category C valued in Set can be defined as the functor $\mathcal{F} : \mathsf{C}^{\mathrm{op}} \to \mathsf{Set}$, but we will only consider sheaves on spaces in this note.

We may show the existence of a presheaf that does not meet the sheaf axiom:

Example. Let $\mathcal{A} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathsf{Ab}$ be the constant presheaf of Abelian groups given in previous examples. We may consider the two disjoint open sets $U_1 \cap U_2 = \varnothing$ and $U = U_1 \sqcup U_2$. Consider the section $a_1 \in A = \mathcal{F}(U_1)$ and $a_2 \in A = \mathcal{F}(U_2)$ which $a_1 \neq a_2$ in A. Since the overlapping set is empty, these two sections vacuously agree on $U_1 \cap U_2$, but there is no such $a \in A$ that the restrictions on different open subsets are different group elements.

and another example of a presheaf that is a sheaf:

Example (Sheaf of Continuous Functions)**.** Consider the presheaf of continuous functions $\mathcal{C}^0_X : \mathrm{Open}(X)^{\mathrm{op}} \to \mathsf{cRing}$ over space X.

Proposition 2.4. *Presheaf of continuous functions $\mathcal{C}^0_X : \mathrm{Open}(X)^{\mathrm{op}} \to \mathsf{cRing}$ is a sheaf.*

Proof. Given continuous function $f_i : U_i \to \mathbb{R}$ such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for any $i, j \in I$. Define the continuous function $f : U \to \mathbb{R}$ by

$$f(x) = f_i(x) \quad \forall x \in U_i.$$

Since f_i and f_j coincide on the overlapping domain, this definition is well-defined. Since for each U_i, f_i is continuous, we know that f is continuous. Suppose there is another $g : U \to \mathbb{R}$ such that for all $i \in I$, $g|_{U_i} = f_i$. Then, for all $x \in U_i$

$$g(x) = f_i(x) = f(x)$$

since $\{U_i\}_{i \in I}$ covers X, $g \equiv f$. This shows the uniqueness. Thus, $\mathcal{C}^0_X$ is a sheaf. □

Although a constant presheaf is not necessarily a sheaf, we have a notion of constant sheaves:

Example (Constant $\mathbb{Z}$-Sheaf)**.** The constant $\mathbb{Z}$-sheaf over space X, denoted as $\mathbb{Z}_X$ is given by

$$\mathbb{Z}_X(U) := \{\text{Locally constant function } U \to \mathbb{Z}\}.$$

Since $\mathbb{Z}$ admits the discrete topology naturally, it is equivalent to consider all continuous maps $U \to \mathbb{Z}$. Note that the sheaf $\mathbb{Z}_X$ is explicitly given by

$$\mathbb{Z}_X(U) \cong \bigoplus_{x \in \Pi_0 U} \mathbb{Z} \cdot x,$$

for any open set $U \subseteq X$.

In the context of vector bundles, we are interested in a specific type of sheaf over topological spaces called "ringed space":

Definition 2.5 (Ringed Space)**.** Let X be some space, $\mathcal{O}_X : \mathrm{Open}(X)^{\mathrm{op}} \to \mathsf{Ring}$ be a sheaf of ring. $(X, \mathcal{O}_X)$ is called a ringed space, and $\mathcal{O}_X$ is called the structural sheaf of the ringed space.

With the structural ring, one can define the sheaf of $\mathcal{O}_X$-module by a sheaf of abelian group $\mathcal{E} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathsf{Ab}$ such that for all open sets $U \subseteq X$, $\mathcal{E}(U)$ is an $\mathcal{O}_X(U)$-module:

$$\mathcal{O}_X(U) \times \mathcal{E}(U) \to \mathcal{E}(U), \quad (f, s) \mapsto f \cdot s,$$

and satisfies the compatibility condition under restriction: If $V \subseteq U$, then $(f \cdot s)|_V = f|_V \cdot s|_V$. Equivalently, the $\mathcal{O}_X$-module sheaf is given by a sheaf of abelian group $\mathcal{E}$ over X with a morphism of sheaves

$$\mu : \mathcal{O}_X \otimes_{\mathbb{Z}_X} \mathcal{E} \to \mathcal{E}, \quad f \otimes s \mapsto f \cdot s$$

where the tensor product is defined to be the sheafification $\mathcal{O}_X \otimes_{\mathbb{Z}_X} \mathcal{E} := \mathcal{P}^\sharp$ of the presheaf $\mathcal{P}$ defined by

$$\mathcal{P}(U) := \mathcal{O}_X(U) \otimes \mathcal{E}(U)$$

since the locally defined tensor product may not meet the sheaf axiom in general, although an $\mathcal{O}_X$-module sheaf is the correct algebraic object on a ringed space, it is still too general to represent a vector bundle. Geometrically, the sheaf of sections of a vector bundle should admit local frames; equivalently, after restricting to sufficiently small open sets, it should look like a finite direct sum of copies of $\mathcal{O}_X$. The following examples show pathologies that must be excluded.

Example. Let $(X, \mathcal{C}^0_X)$ be some ringed space associated to some topological space X. Suppose $f \in \mathcal{C}^0_X(X) = C^0(X)$, and let the set of zeros be $Z(f) := f^{-1}(0)$. Assume $Z(f) \cap \operatorname{supp} f \neq \varnothing$ and defined the $\mathcal{C}^0_X$-module sheaf by the exact sequence:

$$\mathcal{C}^0_X \xrightarrow{f\cdot(-)} \mathcal{C}^0_X \longrightarrow \mathcal{F} \longrightarrow 0.$$

Equivalently,

$$\mathcal{F} := \operatorname{coker}\left(\mathcal{C}^0_X \xrightarrow{f\cdot(-)} \mathcal{C}^0_X\right) = \left(U \mapsto \mathcal{C}^0_X(U)/(f|_U)\right)^\sharp.$$

Then $\mathcal{F}$ cannot be modeled locally by the free sheaf $(\mathcal{C}^0_U)^{\oplus r}$ for open sets $U \subseteq X$.

Proof. Indeed, for any zero $x \in Z$ and its open neighborhood $x \in U$, take open subset

$$V \subset U \cap \operatorname{supp} f.$$

Then, since $f|_V$ is non-vanishing, locally the sheaf $\mathcal{F}|_V = 0$ is trivial. Suppose there is an isomorphism $\mathcal{F}|_U \cong (\mathcal{C}^0_U)^{\otimes r}$, then the restriction leads to

$$(\mathcal{C}^0_U)^{\otimes r} \cong \mathcal{F}|_V = 0$$

and this forces $r = 0$ globally on U. Hence, $\mathcal{F}|_U = 0$. But this is impossible: The class of 1 is nonzero on every neighborhood of p, since an equality $1 = f \cdot g \sim 0$ near p would implies at point p

$$1 = f(p) \cdot g(p) = 0 \cdot g(p) = 0.$$

Thus, $\mathcal{F}|_U$ cannot be the zero sheaf, which leads to a contradiction and shows $\mathcal{F}$ cannot be a sheaf of any vector bundle. □

Example (Skyscraper Sheaf)**.** Let X be some space with $p \in X$ non-isolated. Define the skyscraper sheaf $\mathbb{R}_p$ by

$$\mathbb{R}_p(U) := \begin{cases} \mathbb{R}, & \text{if } p \in U, \\ 0, & \text{otherwise.} \end{cases}$$

with the natural restriction maps. It becomes a $\mathcal{C}^0_X$-module sheaf by

$$h \cdot a := h(p)a, \quad \forall h \in C^0(U),\ a \in \mathbb{R}_p(U)$$

whenever $p \in U$, and by the zero action when $p \in U$. $\mathbb{R}_p$ also cannot locally look like a finite free module.

Proof. For every neighborhood $p \in U$ which contains a nonempty open subset $V \subset U \setminus \{p\}$. On such a V, $\mathbb{R}_p|_V = 0$. If we have an isomorphism

$$\mathbb{R}_p|_U \cong (\mathcal{C}^0_U)^{\oplus r},$$

then, restricting the sheaf to V shows that $r = 0$, which contradicting $\mathbb{R}_p(U) = \mathbb{R}$ for every U containing p. Therefore, $\mathbb{R}_p$ is an $\mathcal{C}^0_X$-module sheaf, but it is not vector-bundle-like. □

These examples suggest that an arbitrary $\mathcal{O}_X$-module sheaf is too flexible. To obtain the sheaf-theoretic analogue of vector bundles, we impose a local freeness condition.

Definition 2.6 (Locally Free Sheaves)**.** Let $(X, \mathcal{O}_X)$ be some ringed space and $\mathcal{E}$ be a sheaf of $\mathcal{O}_X$-modules:

1. $\mathcal{E}$ is locally free if there exists an open cover $\mathcal{U} := \{U_i\}$ if X such that there is isomorphism of $\mathcal{O}_X|_{U_i}$-module sheaves
$$\mathcal{E}|_{U_i} \cong \bigoplus_{i\in I} \mathcal{O}_X|_{U_i}$$
2. Locally free sheaf $\mathcal{E}$ is finite locally free if the index sets I is finite.
3. Finite locally free sheaf $\mathcal{E}$ has rank r if the index set has cardinality $|I| = r$.

Intuitively, the locally-freeness is just describing a sheaf of $\mathcal{O}_X$-module that is locally trivialized on some open cover, i.e., morally

Locally free $\mathcal{O}_X$-module sheaves with finite rank = finite-rank vector bundles on $(X, \mathcal{O}_X)$.

More precisely, a vector bundle can be theoretically defined as a section:

Definition 2.7 (Sheaf of Sections)**.** Let rank r vector bundle η over X be given by the projection $\pi : E \to B$, then the corresponding sheaf of sections $\mathcal{E}$ is defined by for all open set $U \subseteq X$,

$$\mathcal{E}(U) := \Gamma(U, E) = \{s \in C^0(U, E) \mid \pi \circ s = \mathrm{id}_U\}$$

In the case that we consider smooth (resp. holomorphic) vector bundles, we can simply change the definition to let the sections be smooth (resp. holomorphic).

As our final topic regards the sheaf theoretic language of vector bundles, we may talk about the transition functions.

Let $\mathcal{E}$ be a rank r locally free $\mathcal{O}_X$-module sheaf over a sufficiently good space X. Take a local trivializing open cover $\mathcal{U} := \{U_i\}_{i\in I}$ with the overlapping region denotes by $U_{ij} := U_i \cap U_j$. Let the local trivialization be a collection of sheaf isomorphisms $\varphi_i : \mathcal{E}|_{U_i} \xrightarrow{\cong} (\mathcal{O}_X)^{\oplus r}$ for $i \in I$,

Definition 2.8 (Transition Function)**.** The transition function of the locally free sheaf (vector bundle) $\mathcal{E}$ on local trivialize cover $\mathcal{U}$ is defined by

$$g_{ij} := \varphi_i \circ \varphi_j^{-1} : (\mathcal{O}_{U_{ij}})^{\oplus r} \xrightarrow{\cong} (\mathcal{O}_{U_{ij}})^{\oplus r}.$$

Note that $g_{ij} \in \mathrm{Aut}_{\mathcal{O}_{U_{ij}}}((\mathcal{O}_{U_{ij}})^{\oplus r}) \cong \mathrm{GL}_r(\mathcal{O}_X)(U_{ij})$ due to the locally freeness.

We can also use ordinary vector bundle language to define it: Let $\phi_i : E|_{U_i} \to U_i \times V$ be the local trivialization on U_i for some $i \in I$,

$$\tau_{ij} := \phi_i \circ \phi_j^{-1} : U_{ij} \times V \to U_{ij} \times V$$

which is given by $\tau_{ij} = \mathrm{id}_{U_{ij}} \times g_{ij}$ for some $g_{ij}(x) \in \mathrm{Aut}(V) = \mathrm{GL}(V)$ for any $x \in U_{ij}$. And, of course, the transition function must coincide with the sheaf axiom, which leads to the following cocycle condition:

Proposition 2.9 (Cocycle Condition)**.** *The transition functions g_{ij} of vector bundle must satisfies the following conditions:*

1. *Identity condition: $g_{ii} = I$.*
2. *Inverse condition: $g_{ij}(x) = g_{ji}(x)^{-1}$ for any $x \in U_{ij}$.*
3. *Cocycle condition: Let $U_{ijk} := U_i \cap U_j \cap U_k$, then for any $x \in U_{ijk}$*
$$g_{ij}(x)g_{jk}(x) = g_{ik}(x)'$$

The proof of this proposition is simply by calculation; we may leave it as an exercise. One can also obtain the transition function of the pullback bundle.

Proposition 2.10. *Suppose $\pi : E \to Y$ leads to a rank r vector bundle over X with transition maps g_{ij} over open cover $\mathcal{U}_Y := \{U_i\}_{i\in I}$, then pullback bundle induced by the continuous map $f : X \to Y$ has transition map*

$$\tilde{g}_{ij} := g_{ij} \circ f : f^{-1}(U_{ij}) \to \mathrm{GL}_r(\mathbb{F})$$

The proof is just to apply the definition of pullback total space.

As stated before, we may prove that vector bundles are locally free sheaves by examining the transition functions.

Proposition 2.11. *Let η be a rank r vector bundle with projection $\pi : E \to X$, then its sheaf of sections is a rank r locally free $\mathcal{C}^0_X$-module sheaf. Conversely, any finite rank free $\mathcal{C}^0_X$-module sheaf leads to a vector bundle*[1].

Proof. ($\Rightarrow$) Since $E \to X$ is a vector bundle, we may find an open neighborhood $U \subseteq X$ for any point $p \in X$ such that $\pi^{-1}U \cong U \times \mathbb{R}^r$. Then, the section on U is given by $s : U \to U \times \mathbb{R}^r$, $s(x) = (x, f(x))$ for some continuous map $f \in \mathcal{C}^0_X(U)$. This gives the isomorphism for

$$\Gamma(U, E) \cong C^0(U, \mathbb{R}^r) \cong C^0(U)^r =: \mathcal{C}^0_X(U)^{\oplus r}.$$

Thus, we have the isomorphism of sheaves over the local trivializing cover that $\mathcal{E}|_U \cong \mathcal{C}^0_X|_{U_i}$.

($\Leftarrow$) Conversely, for locally-free $\mathcal{C}^0_X$-module sheaf $\mathcal{F}$ with rank r, then let $\mathcal{U} := \{U_i\}_{i\in I}$ be the open cover such that $\varphi_i : (\mathcal{C}^0_X|_{U_i})^{\oplus r} \to \mathcal{F}|_{U_i}$ is an isomorphism, then the section of $s \in \mathcal{F}(U_i)$ can be uniquely expressed by

$$s = f_1 e_{i,1} + \cdots + f_r e_{i,r},$$

for some $f_1, \ldots, f_r \in \mathcal{C}^0_X|_{U_i}$. By the definition of a locally free sheaf, on an overlapping set $U_{ij} := U_i \cap U_j$, the isomorphism of $\mathcal{C}^0_X$-module sheaves $\varphi_i|_{U_{ij}}$ and $\varphi_j|_{U_{ij}}$ leads to a transition function

$$g_{ij} := \varphi_i \circ \varphi_j^{-1} \in \mathrm{Aut}_{\mathcal{C}^0_X}\left((\mathcal{C}^0_X|_{U_{ij}})^{\oplus r}\right) \cong C^0(U_{ij}, \mathrm{GL}_r(\mathbb{R}))$$

and based on this fact, one can construct the total space of the vector bundle as

$$E_{\mathcal{F}} := \Big(\coprod_i U_i \times \mathbb{R}^r\Big)\Big/ \sim, \qquad \text{where } (x, v)_j \sim (x, g_{ij}(x)v)_i$$

and it is not hard to check that $\pi|_{U_{ij}} : E_{\mathcal{F}}|_{U_{ij}} \to U_{ij}$ extends to a global map via gluing, and $E \to X$ defined a rank r vector bundle. □

Remark. Note that in the proof above, we made a little notation abuse since there are two ways to define the transition maps. Let $\varphi_i : E|_{U_i} \xrightarrow{\cong} U_i \times \mathbb{F}^n$ be a local trivialization, then the first way is to define the transition map as

$$\tau_{ij} := \varphi_i \circ \varphi_j^{-1}|_{U_{ij}\times\mathbb{F}^n} : U_{ij} \times \mathbb{F}^n \xrightarrow{\cong} U_{ij} \times \mathbb{F}^n.$$

This definition includes a tautological map $\mathrm{id}_{U_{ij}}$ since base points are preserved in the change of coordinates. Another way is to only consider the nontrivial part of the τ_{ij} above on the fiber:

$$g_{ij} : U_{ij} \to \mathrm{GL}_n(\mathbb{F}), \quad \tau_{ij}(x, v) = (x, g_{ij}(x)v).$$

This definition is more straightforward from the perspective of sheaves. It would be clear that these two definitions are equivalent, and the definition we use depends on the context.

[1] We assume the function to be real valued over a sufficiently good space.

In Section 3, we will further discuss how transition maps are determined (up to some equivalence) vector bundles, or, principal $\mathrm{GL}_n(\mathbb{F})$-bundles, via the notion of Čech cocycles.

To construct new vector bundles from existing onesin addition to the pullback bundleone can perform operations analogous to those on vector spaces: direct sums (Whitney sums), tensor products, duals, and tensor powers (including exterior and symmetric powers). All of these constructions are natural in the sheaf theoretic aspect, but we may also list them explicitly here: Let $\mathcal{E}$ and $\mathcal{F}$ be some finite rank locally free $\mathcal{O}_X$-sheaves over space X with rank r and s.

1. Direct sum (Whitney sum): The direct sum $\mathcal{E} \oplus \mathcal{F}$ is defined on each open set
$$(\mathcal{E} \oplus \mathcal{F})(U) := \mathcal{E}(U) \oplus \mathcal{F}(U).$$
2. Tensor product: Defined tensor product over $\mathcal{O}_X$-module sheaves is much harder than defined direct sum, since in general, the tensor product defined on each open set is in general a presheaf instead of a sheaf. It is often more convenient to defined the tensor product as pointwise operation (on stalks)[2] instead of sectionwise[3]:
$$(\mathcal{E} \otimes \mathcal{F})_x := \mathcal{E}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{F}_x.$$
Or equivalently, in the language of sheaf, the tensor product is defined via the sheafification of the presheaf defined scetionwisly: $\mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{F} = \big(U \mapsto \mathcal{E}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{F}(U)\big)^\sharp$.
3. Hom bundle and dual bundle: The Hom bundle is given by the sheaf $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F})(U) := \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E}|_U, \mathcal{F}_U)$, this is still a finite rank locally free sheaf since $\mathcal{E}$ and $\mathcal{F}$ are both finite rank. In particular, the dual bundle is given by $\mathcal{E}^\vee := \mathcal{H}om(\mathcal{E}, \mathbb{F})$ for the base field $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ of the bundle. Also, the Hom bundle can be written as
$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F}) \cong \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{E}^\vee$$
just like vector spaces.

Note that pullback commutes with these constructions, and one can expect the transition function to change in the most natural way in all constructions above.

From now on, vector bundles will be treated primarily as locally trivial bundles or equivalently through their associated principal frame bundles. The sheaf of sections will be used only when sections, local coefficients, or gluing arguments are explicitly involved.

3 Principal G-Bundle and Reduction

In the previous section, transition functions appeared as the local gluing data of vector bundles. We now reinterpret this gluing data from a more intrinsic point of view. The sheaf of local sections of a vector bundle records its local triviality, while the changes between local frames are governed by a structural group such as $\mathrm{GL}_n(\mathbb{F})$. Passing from a vector bundle to its bundle of frames isolates this group-theoretic symmetry and leads naturally to the notion of a principal bundle. This perspective further suggests a universal classification problem: instead of describing bundles by transition functions on each open cover, one seeks a space BG whose maps from X classify principal G-bundles over X. We may start from the concept of a principal G-bundle.

Before we start the section, we need to make another convention that all groups G are pointed locally compact Hausdorff topological groups without further comments.

[2] The fiber is given by $\pi^{-1}(p) \cong \mathcal{E}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$, where $\kappa(x) := \mathcal{O}_{X,x}/\mathfrak{m}_x$ is the stalk of structure ring at x quotient the maximal ideal of function germs vanishes at x.

[3] Similarly, we can use sheafification to construct quotient bundle $\mathcal{E}/\mathcal{F}$ if $\mathcal{F}$ is a subsheaf of $\mathcal{E}$, and the quotient of tensor product bundle leads to exterior power bundle $\Lambda^k\mathcal{E}$.

Principal G-Bundles and Structure Groups. The guiding example is the frame bundle of a vector bundle. Its transition functions are precisely the same functions that glue the original vector bundle, but now they are understood as G-valued ech cocycles describing how local trivial principal G-spaces are glued together.

Definition 3.1 (Principal G-Bundle)**.** A principal G-bundle over space X is the sequence $G \hookrightarrow P \xrightarrow{\pi} X$, with the total space P be a right G-set and the projection $\pi : P \to X$ is surjective, such that

1. $\pi(p) = \pi(pg)$ for all $p \in P$ and $g \in G$.
2. (Local trivialization) For every $x \in X$, there exists an open neighborhood $U \subseteq X$ with a G-equivariant homeomorphism
$$\varphi : \pi^{-1}(U) \xrightarrow{\cong} U \times G$$
with G act on $U \times G$ by
$$(x, h) \cdot g = (x, hg) \ \forall g, h \in G.$$

The group G is called the structure group of the principal bundle.

Note that the local trivialization condition ensures that the action on G on each fiber $P_x := \pi^{-1}(x)$ is freely and transitively since the restriction of local trivialization

$$\varphi|_{P_x} : P_x \xrightarrow{\cong} \{x\} \times G$$

ensures the group action locally behaves as the right multiplication on the group G. Also, the G-equivariance of local trivialization actually implies the first condition that the projection $\pi : P \to X$ is G-invariant. The bundle map and bundle isomorphism of a principal G-bundle is defined in the most natural way, which requires G-equivariant and the following diagram commutes:

$$\begin{tikzcd} P_0 \arrow[r, "\hat{f}"] \arrow[d, "\pi_0"'] & P_1 \arrow[d, "\pi_1"] \\ X_0 \arrow[r, "f"] & X_1 \end{tikzcd} \qquad \text{and} \qquad \hat{f}(pg) = \hat{f}(p)g \ \forall p \in P_0 \ \forall g \in G.$$

There is also a natural pullback of the principal bundle, defined as what one may expect: For $f : X \to Y$ and a principal G-bundle given by $\pi : P \to Y$

$$f^*P := \{(x, p) \in X \times P \mid f(x) = \pi(p)\},$$

with the projection defined by $\pi := \mathrm{pr}_1 |_{f^*P}$.
Similar to Definition 1.10, we can defined the moduli functor of principal G-bundles:

Definition 3.2 (Moduli Functor of Principal G-Bundles)**.** Define the moduli functor of principal G-bundles $\mathrm{Bun}_G : \mathsf{Top}^{\mathrm{op}} \to \mathsf{Grpd}$ by sending

1. Space X to the groupoid $\mathrm{Bun}_G(X)$, with the object be principal G-bundles over X and morphism be bundle isomorphisms.
2. Continuous map $f : X \to Y$ to the induced pullback functor $f^* : \mathrm{Bun}_G(Y) \to \mathrm{Bun}_G(X)$.

To ensure the well-definedness, we need to check the functoriality of pullback.

Proposition 3.3 (Functoriality of Pullback Principal Bundles)**.** *Let X, Y, and Z be spaces; and consider principal G-bundle over Z given by $G \hookrightarrow E \xrightarrow{\pi} Z$. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be continuous maps. Then one can find a natural isomorphism of pullback bundles $f^*g^*E \cong (g \circ f)^*E$.*

The proof is exactly the same as Proposition 1.9, and we may leave it as an exercise.
Given a principal G-bundle over a base space X, there is a canonical way to generate some vector bundles with the same structure group via the linear representation of G. For the sake of maximal generality, we

may consider the general notion of an associate bundle, which is somehow similar to the "tensor product" of a left G-set and a right G-set.

Definition 3.4 (Associated Bundle)**.** Let $G \hookrightarrow P \xrightarrow{\pi} X$ be principal G-bundle, and F be a left G-set. Then the associate bundle is defined by

$$P \times_G F := (P \times F)/G$$

where the group action is given by $(p, f) \cdot g = (pg, g^{-1}f)$. We can equip this space with a natural projection $[p, f] \mapsto \pi(p)$ whose fiber is F, which leads to the associated bundle $F \hookrightarrow P \times_G F \to X$.

In particular, if $G \hookrightarrow P \to X$ is a principal G-bundle, $\rho : G \to \mathrm{GL}(V)$ is a linear representation. Then, one can get a vector bundle with the total space given by

$$E := P \times_G V = (P \times V)/G.$$

This is an associate bundle defined by the G-action given by

$$(p, v) \cdot g = (pg, \rho(g)v),$$

and each fiber of $E \to X$ is isomorphic to V. We may also prove the claim that the fiber of associate bundle $E := P \times_G F \to X$ is F:

Proof of the fiber of associate bundle. To prove the claim explicitly, we may construct a local trivialization of the associated bundle using the local section $s_U : U \to P$. Equivalently, with this choice of local section, there is a corresponding local trivialization

$$U \times G \xrightarrow{\cong} P|_U, \qquad (x, g) \mapsto s_U(x)g.$$

By the requirement that the action $P_x \curvearrowleft G$ is free and transitive, this is indeed an isomorphism, and for any $x \in U$ and $p \in P_x$, there exists $g \in G$ such that $p = s_U(x)g$. Defined the map

$$\psi_U : (P \times_G F)|_U \to U \times F, \qquad [s_U(x)g, f] \mapsto (x, gf).$$

The well-definedness is straightforward since the equivalence relation requires only $(pg, f) \sim (p, gf)$, which ensures the base point $\pi(p) = \pi(pg) = x$ is well defined. Thus the map we construct is well-defined, i.e., $\psi_U(s_U(x)g, f) = \psi_U(s_U(x), gf) = (x, gf)$.
It remains to prove that ψ_U is a homeomorphism. Note that the map is bijective since we can construct the explicit inverse

$$\theta_U : U \times F \to (P \times_G F)|_U, \qquad (x, f) \mapsto [s_U(x), f].$$

One can check that this is indeed the inverse of ψ_U:

$$\begin{aligned} (\theta_U \circ \psi_U)([s_U(x), f]) &= \theta_U(x, f) = [s_U(x), f], \\ (\psi_U \circ \theta_U)(x, f) &= \psi_U([s_U(x), f]) = (x, f). \end{aligned}$$

Finally, we may show that both ψ_U and $\theta_U = \psi_U^{-1}$ are continuous. Notice that both maps factor through the following diagram:

$$\begin{tikzcd}
P|_U \times F \arrow[d, "q"'] \arrow[dr, shift right, "\widetilde{\psi}_U"'] & \\
(P \times_G F)|_U \arrow[r, shift left, "\psi_U"] & U \times F \arrow[l, shift left, "\theta_U"] \arrow[ul, shift right, "(s_U,\mathrm{id})"']
\end{tikzcd}$$

where $$\begin{aligned} \widetilde{\psi}_U : P|_U \times F &\to U \times F \\ (p, f) &\mapsto (\pi(p), \tau_U(p) \cdot f) \end{aligned}$$
and $q : P|_U \times F \to (P \times_G F)|_U$ is the quotient map.

By the definition of principal G-bundle, for every $p \in P|_U$, there is unique $g = \tau_U(p) \in G$ such that

$$s_U(\pi(p)) = p \cdot \tau_U(p).$$

By the continuity of s_U, we know that $\tau_U : U \to G$ is continuous, and thus, $\widetilde{\psi}_U(p, f) = (\pi(p), \tau_U(p) \cdot f)$ is a well-defined continuous map. The universal property of the quotient ensures that ψ_U is continuous, and since $\theta_U = q \circ (s_U, \mathrm{id})$ is the composition of a continuous map, we know that both ψ_U and $\theta_U = \psi_U^{-1}$ are continuous. Thus, ψ_U is a homeomorphism, which completes the proof. □

Furthermore, the following proposition shows that the construction of associate bundle is also functorial:

Proposition 3.5. *Let $P_0 \to X$ and $P_1 \to X$ be principal G-bundles, and $F : P_0 \xrightarrow{\cong} P_1$ is an isomorphism of principal G-bundles, and $G \curvearrowright F$ be a left G-set. Then, the map*

$$F \times_G \mathrm{id}_F : P_0 \times_G F \xrightarrow{\cong} P_1 \times F, \qquad [p, f] \mapsto [F(p), f]$$

is an isomorphism between F-bundles[4].

The proof is simply apply the definition of the map and the G-equivariance of F to show the well-definedness. We may leave the details as an exercise.

The most important example here is the relation between a vector bundle and its frame bundle:

Example. Let $\mathbb{F}^n \cong V \hookrightarrow E \to X$ be a rank n vector bundle over field $\mathbb{F}$, its frame bundle is defined by $\mathrm{Fr}_n(E) \to X$ with total space (on set level) given by

$$\mathrm{Fr}_n(E) = \coprod_{x \in X} \mathrm{Fr}_n(E_x) = \coprod_{x \in X} \mathrm{Isom}(V, E_x).$$

This is a principal $\mathrm{GL}_n(\mathbb{F})$-bundle induced by the the natural right action $f \cdot A = f \circ A : V \to E_x$, where $A \in \mathrm{GL}_n(\mathbb{F}) = \mathrm{Aut}(V)$. It is not hard to check that this action is free and transitive on each fiber, and the following lemma shows the fiber of the frame bundle is indeed isomorphic to G:

Lemma 3.6. *If $(S, s_0) \curvearrowleft G$ is a right-based G-set that G acts freely transitively on S, then with the choice of base point, $G \cong S$ canonically as a right G-set.*

Before we start the proof, we may first state the standard right G-action on $S \times S$ and $S \times G$: For any $(s, g) \in S \times G$ and $(s_1, s_2) \in S \times S$

$$(s, g) \cdot h = (s, gh), \quad (s_1, s_2) \cdot h = (s_1, s_2 h) \quad \forall h \in G.$$

Then, we may prove the lemma:

Proof. Suppose G be a group and S be a based right G-set with a based point s_0 and a free, transitive group action

$$\rho : S \times G \to S, \qquad (s, g) \mapsto sg.$$

Then, it is sufficient to prove the map

$$f := (\mathrm{pr}_1, \rho) : S \times G \to S \times S, \qquad (s, g) \mapsto (s, sg)$$

is a isomorphism of G-sets. The bijectiveness is given by the freeness and transitiveness of the group action:

1. (Free ⇒ injective) Suppose $(s_1, s_1 g_1) = (s_2, s_2 g_2)$, then $s_1 = s_2 = s$ and $sg = sh$. But the second equation is just $s = shg^{-1} = sgh^{-1}$, which implies $g = h$, i.e., f is injective.

[4] Fiber bundle with fiber F.

2. (Transitive $\Rightarrow$ surjective) For any $(s_1, s_2) \in S \times S$, since the group action is transitive, one can find $g \in G$ such that $s_2 = s_1 g$. Thus, $f(s_1, g) = (s_1, s_2)$ which proves the surjectiveness.

Then we need to show that this map is G-equivariant: For any $s \in S$ and $g, h \in G$

$$f(s,g)\cdot h = (s, sg)\cdot h = (s, sgh) = f(s, gh) = f((s,g)\cdot h).$$

Thus, f is a G-set isomorphism.

Finally, consider the based map $\Phi_{s_0} : (G, e) \to (S, s_0)$ given by

$$\begin{array}{ccccccc} G & \xhookrightarrow{i_{s_0}} & S \times G & \xrightarrow[\cong]{f} & S \times S & \xrightarrow{\mathrm{pr}_2} & S \\ g & \longmapsto & (s_0, g) & \longmapsto & (s_0, s_0 g) & \longmapsto & s_0 g \end{array}$$

Note that $i_{s_0}(G) = \{s_0\} \times G$ and $f|_{\{s_0\}\times G} : \{s_0\} \times G \to \{s_0\} \times S$ is still bijective. Since the inclusion map i_{s_0} induces isomorphism $G \cong \{s_0\} \times G$ and pr_2 induces isomorphism $S \cong \{s_0\} \times S$, the map $\Phi_{s_0} : G \to S$ is a set-theoretic bijection given by $\Phi_{s_0} = \mathrm{pr}_2 \circ (f|_{\{s_0 \times G\}}) \circ i_{s_0}$, and the G-equivariant is follows from the equivariance of f. □

Note that the isomorphism is canonical only relative to a choice of base point. Furthermore, in the context of topological group actions, if the action $X \curvearrowleft G$ is free, transitive, and proper, then the map $G \xrightarrow{\cong} X$ induces a homeomorphism. In the case of vector bundle and its frame bundle, things is a little bit easier since the local trivialization provides a construction of continuous inverse: Fix the base point $e_0 \in \mathrm{Fr}_n(E_x)$, the lemma gives $\mathrm{GL}_n(\mathbb{F})$-equivariant[5] bijection

$$\varphi_{x,e_0} : \mathrm{GL}(V) \to \mathrm{Fr}_n(E_x), \qquad A \mapsto e_0 \circ A.$$

We shall use the local triviality of the vector bundle $E \to X$ to construct the continuous inverse. Let $U \subseteq X$ be an arbitrary local trivialization subset with a trivialization

$$\tau_U : E|_U \xrightarrow{\cong} U \times V$$

with the induced fiberwise isomorphism $\tau_{U,x} : E_x \to \{x\} \times V \cong V$ for all $x \in U$. Note that the inverse of this map is a frame $e_{U,x} := \tau_{U,x}^{-1} : V \to E_x$ that continuously depends on x. A natural idea is to construct the map

$$\phi_x : \mathrm{Fr}_n(E_x) \to \mathrm{GL}(V), \qquad f \mapsto \tau_{U,x} \circ f.$$

This local trivialization construction naturally gives a base point $e_U(x) \in \mathrm{Fr}_n(E_x)$, we may check that under this choice of base point, ϕ_x is the inverse of φ_{x,e_U}: Given arbitrary frame $f : V \to E_x$ and linear isomorphism $A \in \mathrm{GL}(V)$

$$\begin{aligned} (\varphi_{x,e_{U,x}} \circ \phi_x)(f) &= e_{U,x} \circ \tau_{U,x} \circ f = (\tau_{U,x}^{-1} \circ \tau_{U,x}) \circ f = f, \\ (\phi_x \circ \varphi_{x,e_{U,x}})(A) &= \tau_{U,x} \circ e_{U,x} \circ A = (\tau_{U,x} \circ \tau_{U,x}^{-1}) \circ A = A. \end{aligned}$$

Thus, $\phi_x = \varphi_{x,e_{U,x}}^{-1}$ provides the continuous inverse of the $\mathrm{GL}(V)$-set isomorphism $\mathrm{GL}(V) \cong \mathrm{Fr}_n(E_x)$, which also shows that $\mathrm{Fr}_n(E_x) \hookrightarrow \mathrm{Fr}_n(E) \to X$ is a principal $\mathrm{GL}(V)$-bundle with local trivialization

$$\mathrm{Fr}_n(\tau_U) : \mathrm{Fr}_n(E)|_U \xrightarrow{\cong} U \times \mathrm{Fr}_n(V) \cong U \times \mathrm{GL}(V).$$

The construction of the frame bundle leads to a functor between groupoids

$$\mathrm{Fr}_n : \mathrm{Vect}_n^{\mathbb{F}}(X) \to \mathrm{Bun}_{\mathrm{GL}_n(\mathbb{F})}(X).$$

[5] Note that $\mathrm{GL}(V) \cong \mathrm{GL}_n(\mathbb{F})$

Conversely, given a principal bundle $\mathrm{GL}(V) \hookrightarrow P \xrightarrow{\pi_P} X$ and the canonical left action $\mathrm{GL}(V) \curvearrowright V$, one can find the associate bundle given by $V \hookrightarrow E \to X$ where the total space is

$$E := P \times_{\mathrm{GL}(V)} V = (P \times V)/\mathrm{GL}(V), \quad (pA, v) \sim (p, Av),$$

with the projection defined by

$$\pi_E : E \to X, \qquad [p, v] \mapsto \pi_P(p).$$

This construction associate bundle gives a vector bundle with fiber V, since we can construct the local trivialization via the section of the principal G-bundle. In this way, one can get another functor

$$(-) \times_{\mathrm{GL}(V)} V : \mathrm{Bun}_{\mathrm{GL}(V)}(X) \to \mathrm{Vect}_n^{\mathbb{F}}(X).$$

Our goal is to prove that this is the natural inverse of the functor Fr_n.
Furthermore, there is natural isomorphism between this associate bundle $\mathrm{Fr}_n(E) \times_{\mathrm{GL}(V)} V \to X$ and the vector bundle $E \to X$:

Proposition 3.7. *There is natural isomorphism* $\mathrm{Fr}_n(-) \times_{\mathrm{GL}(V)} V \cong \mathrm{id}_{\mathrm{Vect}_n^{\mathbb{F}}(X)}$

Proof. Let $\eta \in \mathrm{Vect}_n^{\mathbb{F}}(X)$ $V \hookrightarrow E \to X$ be a vector bundle over X, pick local trivialization $\tau_U : E|_U \xrightarrow{\cong} U \times V$. Then, over each point $x \in U$, the inverse

$$e_{U,x} := \tau_U^{-1}|_{\{x\}\times V} : \{x\} \times V \cong V \xrightarrow{\cong} E_x$$

induces a frame in $\mathrm{Fr}_n(E_x) := \mathrm{Isom}(V, E_x)$, and further leads to the local trivialization of the frame bundle

$$\mathrm{Fr}_n(E)|_U \xrightarrow{\cong} U \times \mathrm{GL}(V), \quad e_{U,x} \circ A \mapsto (x, A).$$

Note that $(\mathrm{Fr}_n(E)\times_{\mathrm{GL}(V)} V)_x = \mathrm{Fr}_n(E_x)\times_{\mathrm{GL}(V)} V \cong V$, consider the bundle map $\Theta_E : \mathrm{Fr}_n(E)\times_{\mathrm{GL}(V)} V \to E$ over X defined fiberwisely as

$$\Theta_{E,x} : \mathrm{Fr}_n(E_x) \times_{\mathrm{GL}(V)} V \to E_x, \qquad [f, v] \mapsto f(v),$$

since $f \in \mathrm{Fr}_n(E_x) = \mathrm{Isom}(V, E_x)$, $\pi_E\big(f(v)\big) = x$, which ensures that the map is a map over X. The well-definedness of this map is obvious.
We shall first show that Θ_E is indeed an isomorphism of vector bundles. Note that the local trivialization of the associated bundle is given by $\phi_U : (\mathrm{Fr}_n(E) \times_{\mathrm{GL}(V)} V)|_U \to U \times V$,

$$[e_{U,x}, Av] \longleftrightarrow (x, Av).$$

Under the local coordinate, the map we defined is given by $\widetilde{\Theta}_{E,U} := \tau_U \circ \Theta_E \circ \phi_U^{-1}$

$$\widetilde{\Theta}_{E,U}(x, v) = (\tau_U \circ \Theta_E)([e_{U,x}, v]) = (\tau_U \circ e_{U,x})(v) = (x, v).$$

Thus, $\widetilde{\Theta}_{E,U} = \mathrm{id}_{U\times V}$ is clearly a homeomorphism, which also implies Θ_E is a homeomorphism.
Finally, we may show the naturality of this construction. Let $\alpha : E \xrightarrow{\cong} E'$ be an isomorphism in $\mathrm{Vect}_n^{\mathbb{F}}(X)$, then there is commutative diagram:

$$\begin{tikzcd}
\mathrm{Fr}_n(E) \times_{\mathrm{GL}(V)} V \arrow[r, "\Theta_E"] \arrow[d, "\mathrm{Fr}_n(\alpha)\times_{\mathrm{GL}(V)}\mathrm{id}_V"'] & E \arrow[d, "\alpha"] \\
\mathrm{Fr}_n(E') \times_{GL(V)} C \arrow[r, "\Theta_{E'}"] & E'
\end{tikzcd}$$

We may check that this diagram indeed commutes:

$$\Theta_{E'}([\mathrm{Fr}_n(\alpha)(f), v]) = \Theta_{E'}([\alpha \circ f, v]) = \alpha(f(v)) = (\alpha \circ \Theta_E)([f, v]).$$

Thus, we have natural isomorphism $\mathrm{Fr}_n(-) \times_{\mathrm{GL}(V)} V \cong \mathrm{id}_{\mathrm{Vect}_n^{\mathbb{F}}(X)}$. $\square$

To see this result more explicitly, note that Θ_E is fiberwisely a linear isomorphism. To show this, note that the frame $e_{U,x} : V \to E_x$ is a picked frame, and since $\mathrm{GL}(V)$ acts freely and transitively on each fiber $\mathrm{Fr}_n(E_x) = \mathrm{Fr}_m(E)_x$, for any $f \in \mathrm{Fr}_n(E_x)$, one can find unique $A \in \mathrm{GL}(V)$ such that $f = e_{U,x} \circ A$. We use the given frame $e_{U,x}$ as the base point, and one can deduce that

$$[f, v] = [e_{U,x} \circ A, v] = [e_{U,x}, Av] = [e_{U,x}, w],$$

where $w := Av$. Under this formalism,

$$\Theta_{E,x} : (\mathrm{Fr}_n(E) \times_{\mathrm{GL}(V)} V)_x \to E_x, \qquad [f, v] = [e_{U,x}, w] \mapsto e_{U,x}(w),$$

and this is a linear isomorphism since $e_{U,x}$ is a linear isomorphism. Using the continuity of Θ_E and Lemma 1.5, we can get an alternative proof of the proposition.

And the inverse direction is given by the following proposition:

Proposition 3.8. *Let* $\mathrm{GL}(V) \hookrightarrow P \to X$ *be a principal* $\mathrm{GL}(V)$*-bundle, then there is a natural isomorphism* $P \cong \mathrm{Fr}_n(P \times_{\mathrm{GL}(V)} V)$*.*

Proof. Let $P \to X$ be $\mathrm{GL}(V)$-bundle, fix $x \in X$ and $p \in P_x$, we can defined a canonical map fiberwisely by

$$\lambda_p : V \to (P \times_{\mathrm{GL}(V)} V)_x, \qquad v \mapsto [p, v],$$

and globally defined $\Lambda_P : P \to \mathrm{Fr}_n(P \times_{\mathrm{GL}(V)} V)$ as $\Lambda_P(p) = \lambda_p$. To ensure this map is well-defined, we need to first show that λ_p is a frame, i.e., a linear isomorphism. It is sufficient to prove that it is injective since the surjectiveness follows from the fact that every element in $(P \times_{\mathrm{GL}(V)} V)_x$ can be written as $[p, v]$ for some $v \in V$. Let $v, w \in V$ be two distinct vectors, then

$$\lambda_p(v) = [p, v] = [p, w] = \lambda_p(w)$$

implies that $v = w$, which proves the injectivity. Thus $\lambda_p \in \mathrm{Fr}_n(P \times_{\mathrm{GL}(V)} V)_x$. To prove this is a bundle map of principal $\mathrm{GL}(V)$-bundles, we also need $\mathrm{GL}(V)$-equivariance. Note that for any $A \in \mathrm{GL}(V)$

$$\lambda_{pA}(v) = [pA, v] = [p, Av] = \lambda_p(Av) \quad \forall v \in V.$$

Thus, $\lambda_{pA} = \lambda_p \circ A$, which shows the $\mathrm{GL}(V)$-equivariance of Λ_P.

It remains to show that Λ_P is a homeomorphism. Consider again the section $s_U : U \to P|_U$ of the principal bundle, and the induced local trivialization

$$P|_U \cong U \times \mathrm{GL}(V), \qquad s_U(x)A \longleftrightarrow (x, A).$$

The associate bundle $E_P := P \times_{\mathrm{GL}(V)} V$ also has local trivialization

$$\psi_U : E_P|_U \xrightarrow{\cong} U \times V, \qquad [s_U(x)A, v] \mapsto (x, Av),$$

which induces the local trivialization of its frame bundle

$$\mathrm{Fr}_n(E_P)|_U \xrightarrow{\cong} U \times \mathrm{GL}(V),$$

which fiberwisely defined by $f \mapsto \psi_{U,x} \circ f$. Now we compute the local coordinate representation of Λ_P, denote by $\widetilde{\Lambda}_P$: Take $p = s_u(x)A \in P_x$

$$\widetilde{\Lambda}_P(x, A) = (x, \psi_{U,x} \circ \lambda_{s_U(x)A}) = (x, \psi_{U,x} \circ \lambda_{s_U(x)} \circ A)$$

where the linear map $\psi_{U,x} \circ \lambda_{s_U(x)A} : V \to V$ is given by

$$(\psi_{U,x} \circ \lambda_{s_U(x)A})(v) = (\psi_{U,x} \circ \lambda_{s_U(x)})(Av) = \psi_{U,x}([s_U(x), Av]) = Av \quad \forall v \in V$$

which is just $A \in \mathrm{GL}(V)$. Thus,

$$\widetilde{\Lambda}_P : U \times G \xrightarrow{\cong} U \times G$$

is just the identity map. This shows that Λ_P is an isomorphism between principal $\mathrm{GL}(V)$-bundles. Finally, we may show the naturality of this construction, which requires checking the following diagram commutes: For $P, P' \in \mathrm{Bun}_G(X)$ and an isomorphism $\beta : P \xrightarrow{\cong} P'$

$$\begin{tikzcd} P \arrow[r, "\Lambda_P"] \arrow[d, "\beta"'] & \mathrm{Fr}_n(P \times_{\mathrm{GL}(V)} V) \arrow[d, "\mathrm{Fr}_n(\beta \times_{\mathrm{GL}(V)} \mathrm{id}_V)"] \\ P' \arrow[r, "\Lambda_{P'}"] & \mathrm{Fr}_n(P' \times_{\mathrm{GL}(V)} V) \end{tikzcd}$$

Note that the induced map over associate bundle is given by

$$\beta \times_{\mathrm{GL}(V)} \mathrm{id}_V : P \times_{\mathrm{GL}(V)} V \to P' \times_{\mathrm{GL}(V)} V, \qquad [p, v] \mapsto [\beta(p), v],$$

and the frame construction is just the composition. Since $\Lambda_P(p) = \lambda_p$, $(\beta \times_{\mathrm{GL}(V)} \mathrm{id}_V) \circ \Lambda_P$ is given by

$$\lambda_p \mapsto (\beta \times_{\mathrm{GL}(V)} \mathrm{id}_V) \circ \lambda_p$$

and for any vector $v \in V$

$$(\beta \times_{\mathrm{GL}(V)} \mathrm{id}_V) \circ \lambda_p(v) = (\beta \times_{\mathrm{GL}(V)} \mathrm{id}_V)([p, v]) = [\beta(p), v] = \Lambda_{P'}(\beta(v)).$$

This completes the proof of the proposition. □

As a summary of all the discussions about the frame bundle above, we can state the main theorem here:

Theorem 3.9. *For a space X, there is an equivalence of groupoids:*

$$\mathrm{Vect}_n^{\mathbb{F}}(X) \simeq \mathrm{Bun}_{\mathrm{GL}_n(\mathbb{F})}(X).$$

Equivalently, one has a bijection between isomorphism classes $\pi_0 \mathrm{Vect}_n^{\mathbb{F}}(X) \cong \pi_0 \mathrm{Bun}_{\mathrm{GL}_n(\mathbb{F})}(X)$.

Finally, we may study the principal G-bundle from the perspective of the transition function. Let $P \xrightarrow{\pi_P} X$ be a principal G-bundle, and $\mathcal{U} := \{U_i\}_{i \in I}$ be a trivializing open cover with $U_{ij} = U_i \cap U_j$ for $i, j \in I$.

Proposition 3.10. *The transition function is given by a collection of G-valued functions $g_{ij} : U_{ij} := U_i \cap U_j \to G$. that satisfies the cocycle condition in Proposition 2.9:*

1. *Identity condition: $g_{ii}(x) = e \in G$ for any $x \in U_i$.*
2. *Inverse condition: $g_{ij}(x) = g_{ji}(x)^{-1}$ for any $x \in U_{ij}$.*
3. *Cocycle condition: Let $U_{ijk} := U_i \cap U_j \cap U_k$, then for any $x \in U_{ijk}$*

$$g_{ij}(x) g_{jk}(x) = g_{ik}(x).$$

Here, the only nontrivial fact we need to remark is the reason that the transition function is G-valued. The transition function is naturally given as the $\mathrm{Aut}_G(P_x)$-valued function since the transition map needs to preserve the base point, and it seems to include both inner and outer automorphisms in the sense of group automorphisms, which is clearly not isomorphic to G in general. However, in the context of principal G-bundles, the automorphism of the fiber is required to be not the automorphism of groups but the automorphism of topological right G-sets, which means here

$$\mathrm{Aut}_G(P_x) := \{G\text{-equivariant homeomorphism } P_x \xrightarrow{\cong} P_x\}.$$

Note that $P_x \cong G$ is a topological group up to some choice of base point in P_x, so we have the following proposition:

Proposition 3.11. *The G-equivariant automorphism of $P_x \cong G$ are left translations, i.e.,* $\mathrm{Aut}_G(G) \cong G$.

Proof. It is obvious that left translations $L_g : G \xrightarrow{\cong} G$ are G-equivariant homeomorphisms, so we may show that any $\varphi \in \mathrm{Aut}_G(P_x)$ is representable by a left multiplication. Since the automorphism is required to be G-equivariant, by the isomorphism $P_x \cong G$

$$\varphi(g_1 g_2) = \varphi(g_1) g_2 \quad \forall g_1, g_2 \in G.$$

Then, let the image of identity be $\varphi(e) = h \in G$, for any $g \in G$, $\varphi(g) = \varphi(e)g = hg$. Thus, any $\varphi \in \mathrm{Aut}_G(P_x)$ is a left translation by $\varphi(e)$, and finally

$$G \xrightarrow{\cong} \mathrm{Aut}_G(G), \qquad g \mapsto L_g$$

is indeed a group isomorphism since $L_{gh} = L_g \circ L_h$ for any $g, h \in G$, which complete the proof that $\mathrm{Aut}_G(G) \cong G$. ☐

It is clear that this result applies to any isomorphism of G-sets with free, transitive right action, we will use this fact in the classification of principal G-bundles.

In this way, we can also see the reason why isomorphism classes of rank n vector bundles over the field $\mathbb{F}$ are uniquely correspond to $\mathrm{GL}_n(\mathbb{F})$-bundles. Since one can establish a standard correspondence (bijective up to some equivalence) between transition functions of such vector bundles and transition functions of $\mathrm{GL}_n(\mathbb{F})$-bundles, the sheaf-theoretic correspondence between vector and principal $\mathrm{GL}_n(\mathbb{F})$-bundles is expected.

[**Derek:** Reduction of structure groups]

Remark. The existence of a bundle metric on a paracompact based vector bundle allows us to make the following convention: For any complex vector bundle, we take the most general structure group to be $\mathrm{U}(n)$; and for a real vector bundle, the most general structure group is $\mathrm{O}(n)$.

4 Classifying Spaces and Universal Bundles

Lecture 5: Classifying Space and Universal Characteristic Classes	**Jul. 12, 2026**

Construction of Classifying Spaces

Universal Bundle

Universal Characteristic Class.

5 Construction of Specific Characteristic Classes

5.1 Thom Class, Orientability, and Euler Class

5.2 Stiefel–Whitney Classes

5.3 Chern Class

5.4 Pontryagin Classes

6 The Obstruction Viewpoints

6.1 Obstruction Theory for Sections of Fibrations

6.2 Postnikov and Whitehead tower

6.3 Obstructions to Frames and Characteristic Classes

7 Thom, Gysin, and Intersection Theory

7.1 Gysin Maps, Thom Isomorphism, and Self-Intersection Formula

7.2 Zero Loci of Sections and Euler Classes

7.3 Degeneracy Loci and Chern Classes

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Appendix A. Language of Category and Functors